

CS 527: Pseudorandomness and Derandomization

Course Code: CS 527

Course Name: Pseudorandomness and Derandomization

L-T-P-C: 3-0-0-3

Prerequisites: CS 208, CS 212, basic discrete probability or equivalent

Intended for: M.Tech / Ph.D. / B.Tech 3rd and 4th year

Mutual Exclusion: None

Approval: 64th BoA (proposed as CS 610)

Preamble

Many algorithmic problems have efficient randomized algorithms, but no known efficient deterministic algorithms. This course studies techniques for derandomizing randomized algorithms through the theory of pseudorandomness.

Course Modules

- **Unit 1: Randomized algorithms:** introduction and motivating examples. (3 hours)
- **Unit 2: Basic derandomization techniques:** the method of conditional expectations, pairwise independence, pairwise independent hash functions. (6 hours)
- **Unit 3: Expander graphs:** vertex and spectral expansion, random walks on expander graphs, explicit construction of expander graphs, application: undirected connectivity in deterministic logspace. (11 hours)
- **Unit 4: Pseudorandom generators:** definition, Nisan-Wigderson generator from average-case hardness, worst-case to average-case reduction. (11 hours)
- **Unit 5: Advanced topics (choose one of the following):** randomness extractors, connection of pseudorandom objects and list-decoding, unconditional pseudorandom generators for weaker computational models, hardness vs randomness. (11 hours)

Laboratory/Practical/Tutorial Modules

None

Text Books

1. Salil P. Vadhan, *Pseudorandomness*, Now Publishers, 2012.
2. Pooya Hatami and William Hoza. *Paradigms for Unconditional Pseudorandom Generators*, Emerald Publishing, 2024.